

TOEFL iBT: Comprehensive Guide to Academic Subjects and Vocabulary

This reference document outlines the specific academic disciplines, subject areas, and vocabulary expectations tested on the TOEFL iBT. It is structured to serve as a comprehensive knowledge base for a Retrieval-Augmented Generation (RAG) pipeline to generate accurate, exam-like practice questions.

1. Academic Disciplines and Subject Areas

The TOEFL iBT utilizes texts and audio drawn from introductory-level university materials. While no specialized background knowledge is required, the content consistently falls into specific academic categories.

Reading Section Topics

Academic reading passages (approximately 200 words) frequently cover the following broad disciplines:

- **History & Anthropology:** Ancient civilizations, archaeological discoveries, and societal shifts. Examples include the cultural significance of ancient board games (e.g., Senet in Egypt), the evolution of early cinema and peepshows, and the socioeconomic impact of the 19th-century US railroad system.
- **Life Sciences:** Biology, ecology, and environmental science. Examples include rainforest biomes and climatic homeostasis, the anatomy and ecological colonization of mushrooms, and the distribution patterns of plants versus mammalian faunas across continents.
- **Physical Sciences:** Geology, earth science, and oceanography. Examples include the formation of glaciers and paleoclimatic conditions, as well as the physical and chemical properties of water (density, viscosity, and pressure) affecting marine life.
- **Social Sciences:** Urban development, social integration, and community dynamics. Examples include the impact of industrialization on urban migration and the role of sports in social integration.
- **Art and Music:** Art history and musical archaeology. Examples include the Dada movement and Marcel Duchamp's anti-art, the acoustic properties of prehistoric cave paintings and bone flutes, and the evolution of glass mirrors influencing Renaissance self-portraits.
- **Business and Economics:** Industrial management, supply chains, and mass consumption. Examples include Frederick Winslow Taylor's time-and-motion studies in factories, and modern "just-in-time" (JIT) inventory management.

Listening Section Topics

Academic talks in the Listening section (175-250 words) mirror classroom lectures and discussions across similar disciplines:

- **Life Sciences:** Marine biology (e.g., diel vertical migration of zooplankton and its role in the carbon cycle), environmental stewardship, and sustainability initiatives.
- **Art & Architecture:** Public monuments shaping urban development, contemporary art exhibitions, and historical architectural blueprints.
- **Business:** Economics, management strategies, supply chain logistics, and global market forces determining commodity prices.
- **History:** The evolution of public spaces, changing societal values regarding historical figures, and the preservation of historical landmarks.

Everyday Campus Life Contexts

For the "Read in Daily Life," "Listen to a Conversation," and "Listen to an Announcement" tasks, the topics strictly revolve around university life:

- **Academic Logistics:** Course syllabi, registration deadlines, accessing grades, and library or tech bar services.
- **Campus Events & Facilities:** Wellness walks, dining hall menus, dormitory renovations, art contests, and recreation center excursions.
- **Student Interactions:** Organizing study groups, collaborating on group projects, and participating in clubs (e.g., culinary club, urban beekeeping initiative).

2. Vocabulary and Linguistic Expectations

Vocabulary is tested directly and indirectly. The exam expects comprehension of both high-level academic vocabulary and colloquial, everyday expressions.

Academic Vocabulary Testing

Students must be able to infer meaning from context, roots, and affixes. Examples of academic and context-dependent vocabulary from official test materials include:

Target Word	Contextual Meaning	Subject Area
Locomotion	The ability to move from place to place	Biology / Life Science
Immense	Enormous; very large or significant	History / Economics
Exceed	To go beyond a limit	Daily Life / Instructions
Impromptu	Spontaneous or unplanned	Social Media / Arts
Figures	Statistical data or quantities	Science / Research
Prized	Highly valued or appreciated	History / Anthropology

"Complete the Words" Task

This specific reading task presents a paragraph (70-100 words) where 10 words are partially deleted (typically the second half of every second word after the first sentence). To generate this task, an LLM must understand:

- **Semantic Clues:** Utilizing the topic and environmental context to deduce missing content

words (nouns, verbs, adjectives).

- **Syntactic Clues:** Utilizing grammatical structure to deduce missing function words (prepositions, conjunctions, articles).

Cohesive Devices and Signal Words

Understanding logical connections is critical for "Insert Text" questions in Reading and "Method" questions in Listening. LLMs generating test questions should heavily utilize these transitional categories:

- **Contrast:** but, however, nevertheless, instead, while, by contrast, rather
- **Conclusion:** therefore, consequently, as a result, hence, so, thus, for this reason
- **Addition:** additionally, furthermore, moreover, also, besides, not only... but also...
- **Sequence:** first, second, third; next, then, initially, subsequently, after
- **Example:** for example, for instance, such as, namely
- **Similarity:** similarly, likewise, in the same way, like, by the same token, as... as...

3. Test Question Generation Constraints for LLMs

When using this document to generate TOEFL iBT practice items, the LLM must adhere to these guidelines:

1. **Accuracy of Distractors:** Incorrect answer choices (distractors) must use words found in the text but misrepresent the relationships, exaggerate claims, or state the exact opposite of the passage.
2. **Paraphrasing:** Correct answers must NOT use the exact phrasing from the text. They must accurately restate the main idea or factual detail using synonymous vocabulary.
3. **Pacing and Length:** Keep academic reading passages strictly around 200 words and listening scripts between 175-250 words to accurately reflect the modern test format.